Primary 5 · Lesson 34 · Student Handout

SSPA Specialty - Calculation Questions Sprint for Full Score

T1 morenumber of digits + T2 decimal + T9 fraction · Three major calculatetrap total attacks · 75 minutes · 1-to-3 Online Lesson

Corresponding textbooks: Calculation questions in the first test paper (accounting for about 40-50% of the paper) · Multi-digit · Decimal · Fraction

Core traps:□ T1 Multi-digit calculation · T2 Decimal operation · T9 Fraction operation - 🔴 SSPA compulsory exam

SSPA related: 🔴 High frequency The average score loss rate for the first calculation question in the scored paper is as high as 35%, mostly due to carelessness rather than ignorance

Prerequisite knowledge:L1–L3 multi-digit · L12–L13 decimal · L7–L10 fraction

|||SEP||| ❶ Multiple digits + zero decimal errors ❷ Automation of common fractions ❸ Comprehensive calculation verification habits❶ Zero mistakes in multi-digit + decimal places ❷ Automated fraction division ❸ Comprehensive calculation verification habits

Student Name: ______ Class: ______ Date: ______ Time Spent: ______

I.calculate warm-up (total 5 questions, 5 minutes)

#	Question	Difficulty	Working Space
W1	Calculate 12,345 + 987,654 = ?		
W2	Calculate 3.4 + 0.56 = ? (Align the decimal points!)		
W3	Calculate LCM(6, 8) = ?		
W4	Calculate 0.6 × 5 = ?		
W5	Divide $\frac{3}{4}$ and $\frac{5}{6}$ (write two equivalent fractions).		

II.Core Knowledge + Worked Examples

Knowledge point 1: Multi-digit + decimal calculation traps (T1 + T2) 🔴 SSPA required exam

The four iron laws of multi-digit calculations:

① **Alignment rules |||SEP|||: Addition and subtraction → Align the same digits (ones digits to ones, tenths to tenths)**Carry mark |||SEP|||: Mark "small 1" on the previous line when adding and carrying, and mark "small dot" when subtracting and borrowing

② **Decimal addition and subtraction |||SEP|||: The decimal points must be aligned! Fill in the missing digits with 0 (for example, 3.4 + 0.56 → 3.40 + 0.56)**Decimal multiplication |||SEP|||: Multiply integers first → Count the total decimal places → Point the decimal point from right to left

③ **Verification habits |||SEP|||: Mental arithmetic estimation (for example, 12,345+987,654 ≈ 120,000 + 990,000 = approximately 1.11 million)**: The decimal points must be aligned! Fill in missing bits with 0 (e.g. 3.4 + 0.56 → 3.40 + 0.56)

④ **Decimal multiplication**: First multiply integers → count the total number of decimal places → click the decimal point from right to left

⑤ **Verification Habits**: Mental arithmetic estimation (for example, 12,345+987,654 ≈ 120,000+990,000 = about 1.11 million)

Example KP1-1: Multi-digit addition (including carry)

Calculation: 2,345,678 + 8,765,432 = ?

❌ Common mistakes

10,110,110

Forgot to add to the previous level when carrying

✅ Correct solution

11,111,110

Bit-by-bit alignment, carry mark at each level, level-by-level inspection

Example KP1-2: Decimal multiplication trap

Calculation: 0.35 × 0.4 = ?

❌ Common mistakes	✅ Correct solution
$0.35 \times 0.4 = 1.40$	$0.35 \times 0.4 = 0.140 = 0.14$
35×4=140, just write 140 and forget the decimal point	35×4=140, 2+1=3 decimal places → 0.140 → the simplest 0.14

🧠 **Tip: "Align the digits for addition and subtraction. Treat as an integer first for multiplication. After counting the decimal places, start from the right point."**

⚠ **T2 most frequent error: 0.3 × 0.4 = 1.2 (forgetting to multiply decimals will get smaller and smaller! Verification: 0.4 times of 0.3 < 0.3, the answer must be < 0.3)**

KP1 Synchronization practice — morenumber of digits + decimal (question 1-10)

#	Question	Difficulty	Working Space
1	3,456,789 + 2,543,211 = ?	🌱	
2	10,000,000 − 3,456,789 = ?	🌱	
3	0.45 + 0.8 = ? (Tip: Align the decimal points)	🌱	
4	0.6 × 0.7 = ?	🌱	
5	5,678,900 − 987,654 = ?	🌿	
6	0.25 × 0.4 = ?	🌿	
7	23,456,789 + 87,654,321 = ?	🌿	
8	0.125 × 0.8 = ?	🌿	
9	99,999,999 + 1 = ? (Test you: Where to carry?)	🌳	
10	0.05 × 0.02 = ? (How many zeros after the decimal point?)	🌳	

PAGE 3: KP2 score calculation trap

Knowledge Point 2: Score Calculation Trap Total Attack (T9) 🔴 SSPA Required Exam

Five-step verification method for fraction calculation:

① **Look at the denominator |||SEP|||: Same? Direct addition and subtraction. different? First common denominator**Find LCM |||SEP|||: Use short division to find the least common multiple (not just find a common multiple!)

② **Common denominator |||SEP|||: Numerator = original numerator × (LCM ÷ original denominator)** ——must be expanded simultaneously

③ **Calculation**: The denominator remains unchanged, the numerators are added and subtracted**Reduction**

④ **: Find the HCF of the numerator and denominator → reduce to the simplest → convert improper fractions into mixed numbers**: The denominator remains unchanged, and the numerators are added and subtracted.

⑤ **reduce**: Find HCF of numerator and denominator → reduce to the simplest → convert improper fractions into mixed numbers

①	②	③	④
Looking for LCM	**common points**	**calculate**	**reduce**
Short division least common multiple	The numerator and denominator change simultaneously to LCM.	Adding and subtracting numerators without changing the denominator	Simplest false fraction→mixed fraction

Example KP2-1: Addition of different denominators

$\frac{3}{8} + \frac{5}{12} = ?$

❌ **Common mistakes**

$\frac{8}{20}$

Numerator + numerator, denominator + denominator: 3+5=8, 8+12=20

✅ **Correct solution**

$\frac{19}{24}$

LCM(8,12)=24 → $\frac{9}{24}+\frac{10}{24}=\frac{19}{24}$

Example KP2-2: Subtraction with mixed numbers

$2\frac{1}{3} - 1\frac{1}{2} = ?$

❌ **Common mistakes**

$2-1=1, \frac{1}{3}-\frac{1}{2}=\frac{1}{6} \to 1\frac{1}{6}$

Subtract whole numbers and fractions separately, but 13−12 should be 26−36=−16

✅ **Correct solution**

$\frac{5}{6}$

Conversion to false fraction: 73−32 → LCM(3,2)=6 → 146−96=56

🧠 **Tip: "If you add points, turn them into false ones first, find LCM through points, and follow the multiplication of the numerator, the answer will definitely be the same."**

KP2 synchronization practice — fractioncalculate (question 11-25)

#	Question	Difficulty	Working Space
11	$\frac{1}{4} + \frac{1}{8} = ?$	🌱	
12	$\frac{1}{3} + \frac{1}{9} = ?$	🌱	
13	$\frac{2}{5} + \frac{1}{3} = ?$	🌿	
14	$\frac{3}{4} - \frac{1}{6} = ?$	🌱	
15	$\frac{5}{6} - \frac{1}{4} = ?$	🌿	
16	$\frac{2}{3} + \frac{3}{5} = ?$	🌿	
17	$1\frac{1}{4} + 2\frac{2}{3} = ?$	🌳	
18	$3\frac{1}{2} - 1\frac{2}{3} = ?$	🌳	
19	$\frac{3}{8} + \frac{5}{12} = ?$	🌿	
20	$\frac{7}{10} - \frac{2}{5} = ?$	🌿	
21	$\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$	🌳	
22	$2\frac{1}{5} + 1\frac{3}{10} = ?$	🌳	
23	$\frac{4}{9} + \frac{2}{3} = ?$	🌿	
24	$1\frac{2}{3} + 2\frac{3}{4} = ?$	🌳	
25	$\frac{8}{15} + \frac{7}{10} = ?$ (LCM(15,10)= ?)	🌳	

PAGE 4: KP3 Comprehensive Calculation Strategy

Knowledge point 3: Comprehensive calculation strategy - verification = full score 🔴 SSPA

Strategies for getting full marks on calculation questions (don' t just complete the calculation, you need to verify!):

① Estimation verification method |||SEP|||: Approximate mental arithmetic value (for example, 2.98 × 4.1 ≈ 3×4 = 12, the answer should be close to 12) Inverse operation verification |||SEP|||: Use subtraction to test addition, and use division to test multiplication

② **Unit check**: Is the answer in the correct unit? Is the decimal point position reasonable?

③ **Mantissa check |||SEP|||: only look at the single digit/last digit, quick verification**Special value alert

④ **mantissa check**: Only look at single digits/last digit, quick verification

⑤ **Special value alert**： ×0.5 = ÷2，÷0.5 = ×2，÷0.1 = ×10

Example question KP3-1: Estimation verification

Calculate 198 × 52 = ? First use the estimate to determine which of the following answers is reasonable: A. 1,029 B. 10,296 C. 102,960

❌ Choose without estimating	✅ Estimate verification
Choose A or C	B. 10,296
No habit of estimating = giving away points to a trap	198≈200, 200×50=10,000 → Only B is close

Example KP3-2: Mixing fractions and decimals

Calculate $\frac{1}{2}$ + 0.25 = ? (Convert to decimals or fractions)

❌ Error	✅ Correct
Add directly: $\frac{1}{2}$+0.25=0.50.25=?	$0.5 + 0.25 = 0.75 = \frac{3}{4}$
Fractions and decimals cannot be added directly!	Convert to decimals: 12=0.5, 0.5+0.25=0.75

⚠ **35% of the points lost for the calculation questions in the first paper were due to "no verification" rather than "not knowing how to calculate". Get into the habit of estimating!**

KP3 comprehensivecalculatepractice (question 26-35)

#	Question	Difficulty	Working Space
26	Estimate: 4,987 × 31 ≈ ? Which one is closest? A. 15,000 B. 155,000 C. 1,550,000	🌱	
27	$\frac{3}{4}$ + 0.5 = ? (Convert to decimal calculation)	🌱	
28	Verify with estimate: 12,345 + 87,655 approximately = ?	🌱	
29	0.75 + $\frac{1}{4}$ =? (Unified score calculation)	🌿	
30	2.5 × 4.2 = ? Use estimates first (2.5×4=10), and then calculate arithmetic.	🌿	

31	Verify using the inverse operation: If 3,456 ÷ 12 = 288, verify using multiplication.	🌿	
32	$\frac{2}{5}$ + 0.6 = ? (Unification: 0.4 + 0.6 = ?)	🌿	
33	Estimate: 0.48 × 9.7 ≈ ? Which one is closest? A. 0.4656 B. 4.656 C. 46.56	🌿	
34	$1\frac{1}{4}$ + 0.75 = ? (1.25 + 0.75 = ?)	🌳	
35	Calculate $1-\frac{1}{3}-0.25$ = ? (Add and subtract one after another, pay attention to the order)	🌳	

Sprint speed-up practice (question 36-45 · time limit 15 minutes)

#	Question	Difficulty	Working Space
36	45,678 + 54,322 = ?	🌱	
37	0.8 × 0.25 = ?	🌱	
38	$\frac{3}{4}+\frac{1}{8}$ = ?	🌱	
39	1,000,000 − 456,789 = ?	🌿	
40	0.06 × 0.5 = ?	🌿	
41	$\frac{5}{6}-\frac{2}{9}$ = ?	🌿	
42	2,345 × 67 = ? (Use estimates to verify the rationality of the answers)	🌳	
43	$\frac{1}{2}+\frac{2}{3}+\frac{3}{4}$=? (Three numbers can be divided into three numbers)	🌳	
44	$0.125+\frac{3}{8}$ = ? ($0.125=\frac{1}{8}$)	🌳	
45	$4\frac{1}{3}+2\frac{5}{6}-1\frac{1}{2}$=? (Mixed addition and subtraction - it is recommended to convert all improper fractions)	⛰️	

III.The Lessoncorecommon errorssummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge🌿Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	trap	Correct Approach
1	**Multi-digit addition and subtraction misaligned**	T1	Units place to units place, tens place to tens place, hierarchical alignment
2	**Carry/Borrow Missing Flag**	T1	Carries are marked with a small number on the upper row, borrows are marked with a small dot on the borrowed number.
3	**Decimal points are not aligned for decimal addition and subtraction**	T2	The decimal points must be aligned, and any missing digits must be padded with 0
4	**Forgetting the decimal point when multiplying decimals**	T2	First multiply integers → count the total number of decimal places → click from right to left
5	**0.3 × 0.4 = 1.2**	T2	The more you multiply a decimal, the smaller it gets: 0.3×0.4=0.12
6	**Direct addition and subtraction of fractions with different denominators**	T9	Must be divided first → LCM → simultaneous expansion of molecules
7	**After the common denominator, only the denominator is changed and the numerator is forgotten.**	T9	Numerator = original numerator × (LCM ÷ original denominator)
8	**The answer is not reduced**	T9	Find HCF → reduce to the simplest → false → mixed fraction
9	**Error in adding and subtracting mixed numbers**	T9	Recommendation: Convert all false scores and then calculate
10	**No verification habit**	T1/T2/T9	Use estimates to determine the reasonableness of answers

Comprehensive method to submit calculation questions for score test 🔴 Must read

Three principles of not doing:

① **Do not do it if it is not aligned |||SEP|||: Calculate multiple digits and decimals, never start counting if the digits are not aligned**Do not do it without common denominator |||SEP|||: Fractions with different denominators, LCM is not found, never start adding or subtracting

② **Do not hand in the paper without verification**: Calculate each question mentally once, if the answer is unreasonable, recalculate immediately

③ **Submission test strategy:**Calculation questions account for 40-50% of Paper 1, which is the easiest part to get full marks! Because there are standard answers, it is not an open question. Achieving "zero mistakes" in calculation questions = half of the success in the sub-test.

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🧠 **Tips for calculating full marks: "Align the bits, divide first, then multiply the points, reduce the tail, and estimate the test."**